Soumik Saha

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.70
Institute of Science & Technology	Class XII (West Bengal State Council of Technical & Vocational Education And Skill Development)	2018	7.90%
Kendriya Vidyalaya IIT Kharagpur Midnapur WB	Class X (Central Board of Secondary Education, Delhi)	2015	6.50%

WORK EXPERIENCE

EdiGlobe | Machine Learning Intern

- Applied Python-based data preprocessing, analysis and machine-learning techniques to practical datasets and problems

Department of Mining Engineering, IIT Kharagpur | Research Intern

- Worked on mining and geotechnical studies involving pit and dump slope stability, geo-mining conditions and mine safety.
- Contributed to studies across coal and limestone mines under Prof. Debashish Chakravarty and Prof. Biswajit Samanta.

Indian Railways | Engineering Intern

- Gained practical exposure to railway engineering systems and technical operations.

AI/ML PROJECTS

Graph RAG

- Built relationship-aware retrieval using ChromaDB, NetworkX, LangChain, Hugging Face and Groq.
- A first RAG pipeline built with LangChain, Gemini embeddings, and Chroma -- loads a text file, splits it into chunks, and answers questions over it.
- Sets up a local ChromaDB collection and a retrieval pipeline with document loaders (PDF, HTML, Excel), spaCy, and a Streamlit front end served via ngrok, using the Groq API for generation.

Enterprise AI Knowledge Hub

- Developed document Q&A using **LangChain**, **ChromaDB** and **Google Gemini** with semantic retrieval and source references.
- A more complete **RAG** system for enterprise document Q&A: multi-format ingestion (PDF, DOCX, TXT, MD, HTML), chunking, a persistent Chroma vector store, and a conversational retrieval chain.

Customer Feedback Topic Modeling

- Converts text into numerical features (**TF-IDF** / Count Vectorizer) and applies Latent Dirichlet Allocation (**LDA**) to discover topics, alongside a transformer-based pipeline.

Covid - 19 Early Case Trend Analysis and Recovery Insights

- Exploratory data analysis of early COVID-19 patient records -- demographics, infection sources, regional spread, and a regression model for recovery trends.

Google GenAI SDK exploration

- Hands-on exploration of the google-genai SDK -- listing available Gemini models, streaming chat with a custom persona, and multimodal (image) prompts.

PRODUCT & BUSINESS PROJECTS

RegTech & Generative AI Compliance Analysis

- Evaluated Cmplai's AI-powered RegTech platform, market opportunity, competitive positioning and compliance automation use cases.

INDUSTRY PROJECTS

Railway Alternator Systems | Indian Railways

- Studied the design, operation and application of alternator systems in Indian Railways, with exposure to railway electrical and power-generation systems.
- Gained practical understanding of alternator applications across locomotives, EMU/DEMU and MEMU rolling stock and their integration with railway electrical systems.

South Khliehjari Limestone Mine Meghalaya

- Assessed slope stability and geo-mining/geotechnical conditions of South Khliehjari Limestone Mine, operated by Meghalaya Cements Limited.

Pit & Dump Slope Stability Analysis | Dhirauli Block

- Evaluated pit and dump slope stability using geological and geo-mining conditions from detailed coal exploration at Dhirauli Block.

RELEVANT COURSES

Machine Learning and Its Applications, Strategic Management, Finance & Accounting for Engineers, Operations Management, Entrepreneurial Marketing, Design Thinking, Introduction to Intellectual Property Rights,

SKILLS

Python, SQL, Node.js, Pandas, NumPy, Scikit-learn, Excel, Machine Learning, Generative AI, NLP, RAG, Graph RAG, LDA, TF-IDF, Sentiment Analysis, LangChain, Gemini, Groq, Hugging Face, ChromaDB, spaCy, NetworkX, Product Strategy, Market Research, Business Models, Technology Commercialization, Design Thinking, Strategic Management, Data Analytics, Optimization, Demand Forecasting, Supply Chain, Operations Management,CVP, Break-even-Analysis

POSITIONS OF RESPONSIBILITY

Outreach & Assessment Cell Coordinator | OCS, IIT Hyderabad

- Support industry outreach, placement coordination and recruitment.

Volunteer | International Relations (IR) Cell, IIT Hyderabad

- Selected as a Volunteer of the International Relations Cell, contributing to activities supporting international engagement and institutional outreach.
- Assisted in coordination, communication, and event-related activities involving international students, visitors, and institutional stakeholders.

MSME Tech Connect 2026 | IIT Hyderabad

Head of Infrastructure & Hospitality

Hospitality & Logistics Coordinator | EML, IIT Hyderabad

- Coordinate logistics and hospitality for entrepreneurship and management activities.